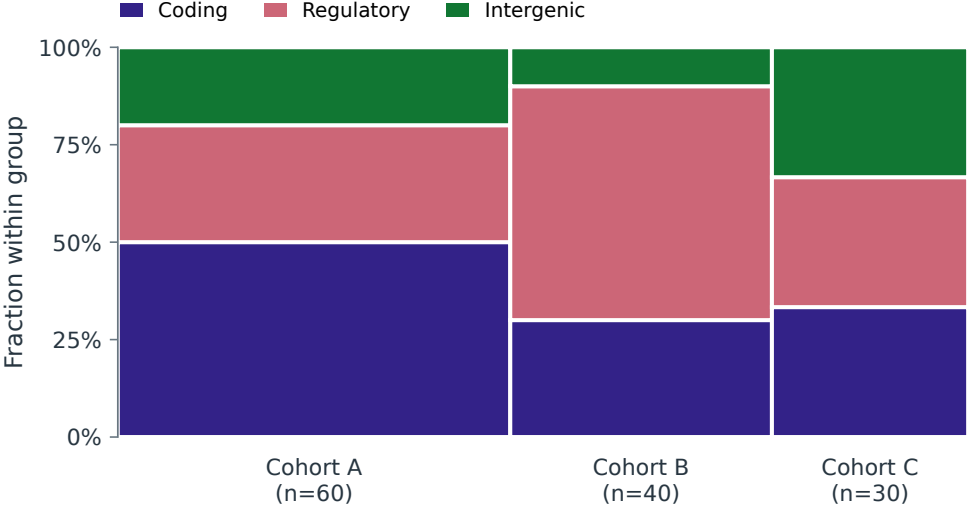Composition and sample size